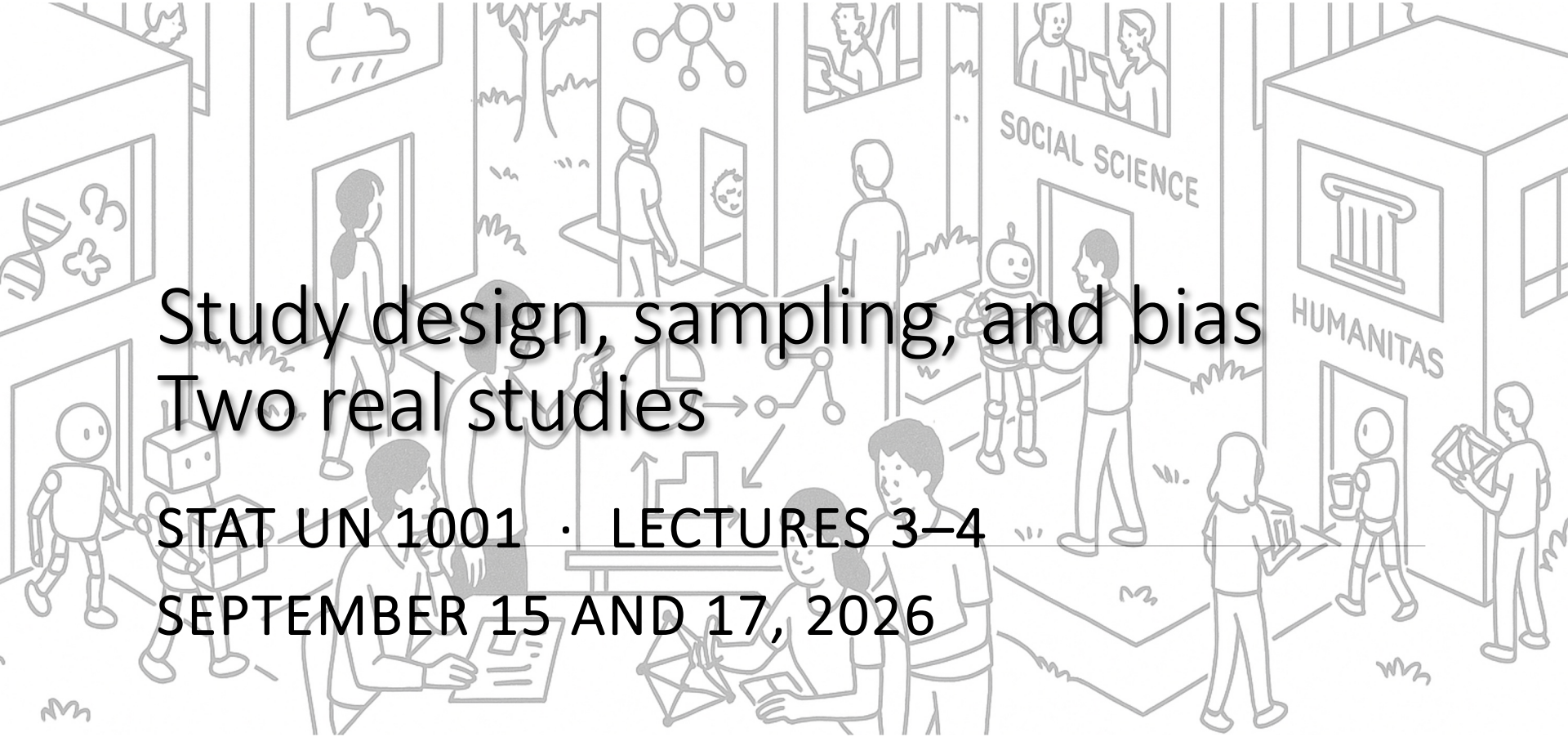


Study design, sampling, and bias
Two real studies

STAT UN 1001 · LECTURES 3–4

SEPTEMBER 15 AND 17, 2026

Random sampling and random assignment

RANDOM SAMPLING Applicability of Results RANDOM ASSIGNMENT Experiments ONLY

Who enters the study?

A probability sample can support generalization from the sample to the population.

Main threat: selection bias

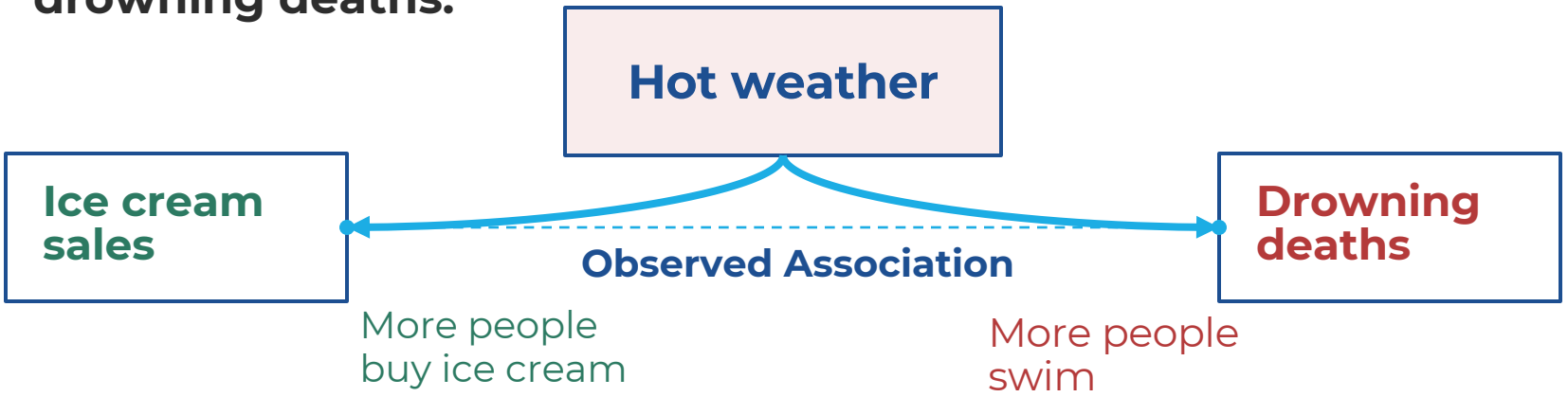
Who receives each treatment?

Random assignment can support a causal comparison among the study participants.

Main threat: confounding

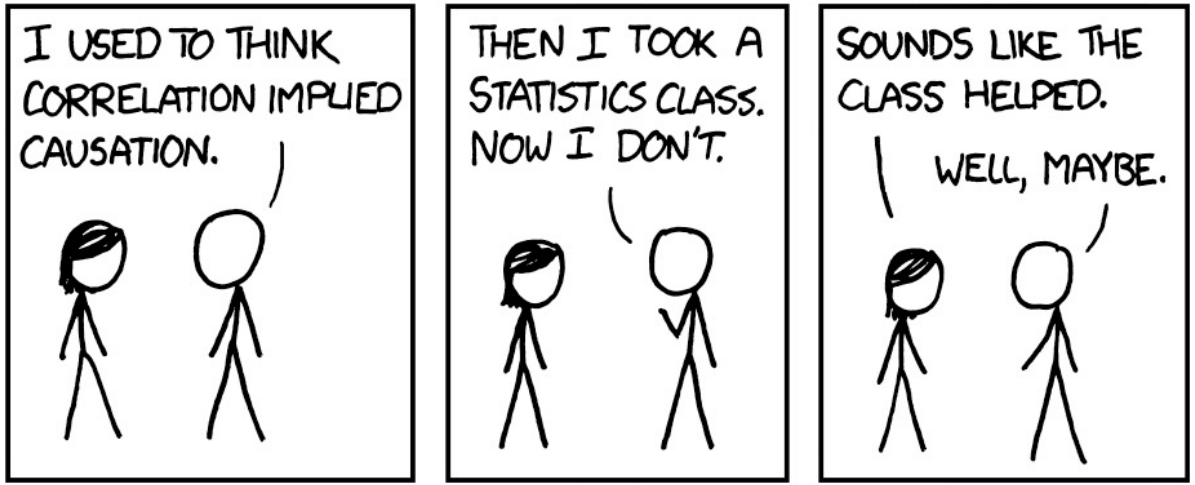
Why Random Assignment?
Example Ice cream sales and drowning deaths

Months with higher ice cream sales also tend to have more drowning deaths.



Correlation

I< < Prev Random Next > >I



I< < Prev Random Next > >I

Permanent link to this comic: https://xkcd.com/552/

Image URL (for hotlinking/embedding): https://imgs.xkcd.com/comics/correlation.png

Why Random Assignment?
Example Practice-tool use and quiz scores

Self-selected comparison

Students choose whether to use the tool.

Motivation can increase both tool use and quiz performance.

A score difference may mix the tool effect with motivation.

Randomized comparison

Researchers assign the tool at random.

Random assignment balances motivation and other background variables on average.

A later score difference supports a causal comparison for the participants.

Blocking by prior quiz score can further reduce chance imbalance in an important known predictor.

Why Random Sample?
How to randomly sample?



Probability sampling designs Individual selection

Simple random sample

Use the full course roster. Select 60 students so every set of 60 has the same chance of selection.

Strength Known and equal inclusion probabilities

Systematic sample

Choose a random starting point on the roster, then select every kth student.

Watch for A roster pattern that matches the sampling interval

Both designs select individual students directly from a complete sampling frame.

Stratified, cluster, and multistage sampling

Stratified sample

Separate students by lecture section or prior performance, then randomly sample within every group.

Purpose: representation and precision

Cluster sample

Randomly select whole recitation sections, then invite every student in the selected sections.

Purpose: lower collection cost

Multistage sample

Randomly select recitation sections, then randomly select students within those sections.

Purpose: balance cost and coverage

Strata all contribute observations. – more balanced representation
Versus
A cluster sample observes only the selected clusters.

The selection rule for today

Does every individual in the population have an equal chance of being selected?

If not, ask which individuals have larger, smaller, or zero chances of entering the data.

Unequal probabilities are most dangerous when they are unknown or related to the variable we want to measure.

News brief Happy adopters still report challenges

SHELTER MEDICINE · FEBRUARY 20, 2026

More than 22,000 adopters from 112 shelters and rescues in 40 states answered surveys during the first month after adoption.

More than 94% rated the adoption positively at each survey time. Yet more than 78% of dog adopters and 50% of cat adopters reported at least one challenge.

Surveys were sent on Day 1, Week 1, and Month 1. A response counted only while the pet remained in the home.

Paraphrased from the University of Florida Shelter Medicine Program

Case 1 Adoption satisfaction

More than

22,000

adopters surveyed

112

shelters and rescues

40

states represented

Over 94% rated the adoption positively at each survey time.

Responses counted only when the pet remained in the home.

Adoption study Population and individuals

PAIR DISCUSSION 4 MINUTES How would you organize the data?

Population	Which larger group do we want the 94% to describe?
Individual	Is one individual an adopter, an adoption, a pet, or a survey response?
Observation (cases)	The same adoption may produce a response on Day 1, Week 1, and Month 1.

Cases					

Adoption study - Who entered the data?

PAIR DISCUSSION 5 MINUTES

1

All adoptions

Target population of shelter and rescue placements

2

Organizations

Only participating shelters and rescues

3

Respondents

Only adopters who answered a survey wave

4

Pet still home

Responses excluded after a return or other departure

At which step do some individuals receive a smaller or zero chance of appearing?

Adoption satisfaction - measurement

Variable of interest

Satisfaction with the adoption experience

Measured value

Rating from 1 to 5

Ratings 4 or 5 were classified as positive

Challenges were measured separately, so an adopter could be satisfied and still report a problem.

Does this operational definition capture everything meant by a successful adoption?

Adoption selection bias and redesign

Why observed satisfaction may shift

Adoptions ending in return or rehoming may have different satisfaction ratings.

Excluding them can make positive experiences more common in the observed data than in the target population.

A probability-based redesign

1. Build a frame of all completed adoptions

2. Randomly sample placements

3. Follow every sampled placement, including returns

4. Record and adjust for nonresponse

Would the redesigned sample estimate a different satisfaction rate? Explain the direction you expect and the assumption behind it.

Redesigning a weak study

A stronger design changes at least one part of the study.

1. How individuals enter the data
2. How the variables are measured
3. How treatments are assigned when causality is the goal

News brief - Homicide fell sharply in 35 cities

PUBLIC SAFETY · JANUARY 22, 2026

A year-end report found that homicide rates fell 21% from 2024 to 2025 across 35 U.S. cities, about 922 fewer homicides.

Thirty-one of the 35 cities recorded declines. The report used cities that consistently supplied monthly police or municipal data.

The authors warned that the cities do not form a nationally representative sample.

Paraphrased from AP News and the Council on Criminal Justice

Homicide study Population and individuals

Population	If the claim is national, which places belong in the target population?
Individual	For selection, is the individual a person, a homicide, or a city?
Observation	A city-year supplies a count and a rate. Two years allow a change to be calculated.

Homicide study Who entered the data?

1

All U.S. cities

Target population implied by a national headline

2

Available systems

Police and municipal records with usable definitions

3

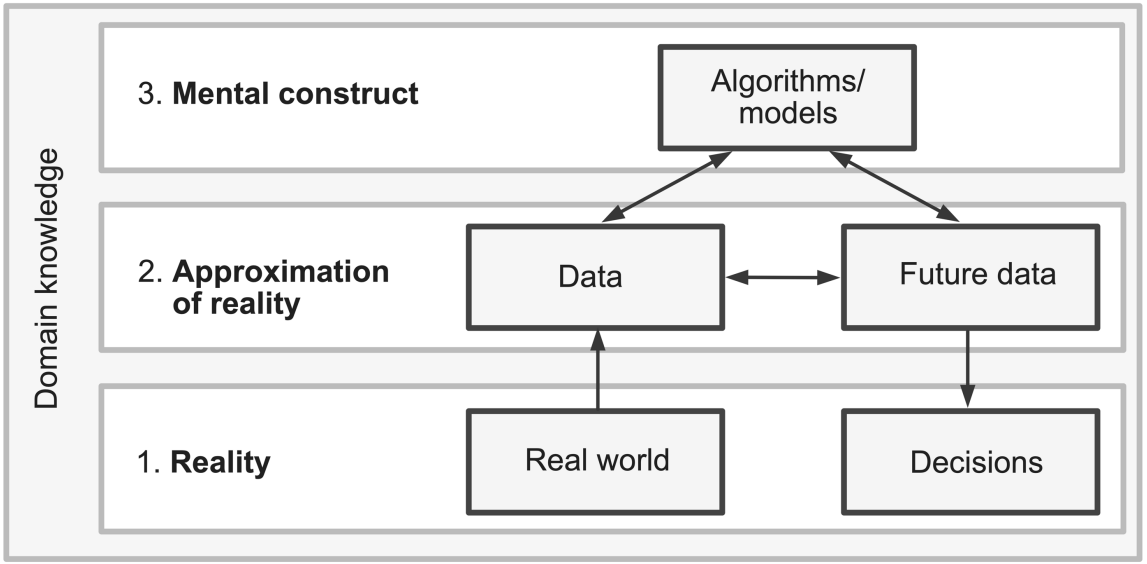
Timely reporters

Cities reporting consistent monthly figures

Final study sample: 35 cities

Which cities had little or no chance of appearing, and how might they differ?

VDS three realms: data and future data



Reality

All individuals and their true values in the target population

Data

The individuals selected and the variables recorded now

Future data

What another sample or later measurement would record

From a study description to a data biography

GENERAL DESCRIPTION

“A survey of adopters found that more than 94% rated their adoption positively.”

A headline tells us the result.

A data biography tells us how the result became possible.

DATA BIOGRAPHY

Who produced the data, and why?

How were observations generated or selected?

What are the cases and variables? How were they measured?

What is covered, missing, filtered, or excluded?

Which claims can these data support?

Let’s find some data with Taffy and Build a first data biography

Mini-project 0: Question and Retrieval Trial

Purpose

Turn an area of interest into a statistical question and test whether the course Data Search and Retrieval Assistant can locate appropriate data.

Most work will be completed during class with instructor and TA support.

In-class work

- Choose weather, environmental sensors, disaster response, or 311 complaints as a domain.
- Write a question with a specific population or process, place, time period, and variables.
- Use the Data Search and Retrieval Assistant to locate a relevant dataset or subset.
- Identify cases, variables, variable types, units, population, and available geographic and temporal coverage.
- State one reason the data may not fully answer the question, then revise the question to match the available data.

Let’s find some data with Taffy and Build a first data biography

Mini-project 1: Data Biography and Retrieval Audit

Purpose

Investigate where retrieved data came from, what they measure, what they omit, and what claims they can support before beginning analysis.

Most work will be completed during class with instructor and TA support.

In-class work

- Continue with Mini-project 0 data or retrieve a more suitable dataset.
- Identify the original data producer, source, and collection or generation process.
- Define cases and variables, including units and whether variables are measured, counted, classified, or constructed.
- Record geographic and temporal coverage, aggregation, missingness, filters, exclusions, and representation concerns.
- Write one claim the data could support and one they could not support.

Submit

- Statistical question, original source, and data producer.
- Collection process, cases, variables, definitions, and units.
- Coverage, aggregation, missingness, exclusions, and representation concerns.
- One supportable claim and one unsupported or overreaching claim.